MAFNet: Deep Merged Autofocusing Network for SAR Recognition Under Defocused and Noised Dataset

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***Abstract*— Airborne synthetic aperture radar (SAR) imagery is susceptible to nonsystematic motion errors, which will definitely defocus the target image and degrade the recognition accordingly. Representations learned from conventional convolutional neural networks (CNNs) do not emphasize much about the intrinsic features of the data distribution, which collapses the purposefulness of feature extraction. To this end, a deep merged autofocusing network (MAFNet) is proposed for adaptively removing the defocusing effect from the input SAR data used for target recognition. Specifically, we propose an end-to-end architecture consisting of one focusing module and one recognition module. The focusing module contains a U-shaped subnetwork based on the cross convolution to ensure the phase history coherence of the extracted features, and a deep unfolding method is devised to improve the mathematical generalization of focusing and sparse features so as to enhance the purposefulness of feature extraction. In particular, a compact surrogate function is designed for the nonconvex feature quantization problem in the focusing module, which leads to closed-form solutions. The recognition module simply consists of a classifier. By building a multivariate loss function consisting of a cross-entropy loss function and a novel focusing loss function, MAFNet achieves accurate target recognition even when the input data are contaminated by nonsystematic phase errors and additive noises. The MSTAR dataset is utilized to validate the effectiveness of MAFNet, and comparisons with conventional algorithms are performed to demonstrate the superiority of the proposed network.**

***Index Terms*— Autofocusing, convolutional neural network (CNN), synthetic aperture radar (SAR), unfolding.**

I. Introduction

SYNTHETIC aperture radar (SAR) is a kind of coherent imagery sensor, and it is capable of generating high-resolution images of stationary scenes and moving targets in all day-and-night and all weathers by utilizing relative movements between the airborne platform and scenes/targets of interests [1], [2], [3], [4], [5]. Motion is not only a necessary condition for generating SAR images but also a key factor affecting image focusing quality. The movement along a predetermined flight track results in a systematic movement, and the SAR image can be formed by classical methods, such as range–Doppler [6] and chip scaling [7], [8]. However, in practice, due to atmospheric turbulence and navigational errors, the nonideal and unknown movements may lead to nonsystematic movements and motion errors, which induces both phase errors and envelope displacements [9]. The errors make the intended SAR image defocused, and the accuracy of the target recognition degraded.

A. Motion Error Compensation

Conventionally, the on-board inertial navigation unit (INU) and global positioning system (GPS) are utilized for error measurement to compensate for nonsystematic motion errors [10]. However, in small-sized and lightweight airborne platforms, such as uncrewed aerial vehicle (UAV), the payload is too small to mount on an accurate INU/GPS. Moreover, as the resolution of SAR imagery increases, the need for high-precision navigation systems for the conventional method increases, but high-precision navigation data are difficult to obtain in practice. The measurement accuracy of INU/GPS is limited because the long-time integration needs to be carried out within the synthetic aperture time, and there is a large accumulation error. In such cases, algorithms based on autofocusing are necessary to estimate nonsystematic motion errors from the echoed data. Commonly, the autofocusing algorithms can be classified into three categories, namely, map drift (MD) [11], phase gradient autofocusing (PGA) [12], and minimum entropy autofocusing (MEA) [13]. The MD algorithm is based on the polynomial fitting model and achieves error compensation by estimating the quadratic error coefficient, which limits the accuracy of error estimation. The PGA algorithm has no restriction on the algorithmic model of the phase error. However, it depends on the presence of dominant scattering points. The MEA algorithm directly estimates the phase error function through the regularization optimization. It is applicable in the low-SNR scenarios, where there would be no dominant scatters. However, the optimization in the MEA algorithm depends on the concavity of the objective function, or no closed-form solutions can be guaranteed.

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B. Feature Quantization

Conventionally, the SAR image formation processing (IFP) [14] algorithms work in the range–Doppler domain under the azimuth invariant assumption to correct range cell migrations, while autofocusing is employed to compensate motion errors, if necessary. Fourier transformation is subsequently applied to achieve the focused images. In such a case, the image response is limited in a Sinc function with fixed main-lobe width and high sidelobes. Therefore, it is difficult to extract detailed features of the target of interest. To this end, compressive sensing (CS) emerges, which suggests that a high-dimensional signal can be recovered from its low-dimensional projection if it is parsimonious [15]. When the SAR imagery meets CS, dominant scatters can be enhanced while the additive interferences will be suppressed. Generally, the sparse-driven SAR imaging algorithms are divided into three categories: including greedy algorithms [16], convex optimization [17], and Bayesian learning [18]. Greedy algorithms [16] are simple but insufficient in high-dimensional problems. The Bayesian learning requires to impose a proper statistical prior on the variable to be solved. It is flexible but problem dependent under specific formulations [18]. Regarding to the accuracy and efficiency, the convex regularization is superior in accessing the solution of sparse features.

C. Data-Driven Deep Learning

The SAR imagery is inherently a typical inverse problem, and the solution can be classified into model-driven and data-driven methods. All three categories mentioned in CS are model-driven methods. In the model-driven methods, the target features are formulated into an analytical model hypothesis, and the problem is solved in either regularization optimization or statistical Bayesian inference. The model-driven learning owns a strong ability of mathematical generalization, where the physical significance of the features can be well-interpreted with clear mathematics. However, as the complexity of the problem increases, it becomes questionable to solve closed-form solutions to practical nonlinear problems. Due to the nonlinearity, the model hypothesis may be mismatched, and the solution would be inaccessible.

Alternatively, the data-driven method is superior in solving nonlinear problems, while large training datasets and accurate supervision labels are required. Deep learning is a typical method of data-driven algorithms [19], [20], [21], which generally employs neural network architecture to fit the training data for an accurate formulation [22], [23], [24], [25]. Convolutional neural network (CNN) exploits multiple layers in a data convolutional manner, so that the convergence efficiency of the training process can be improved [26]. CNN performs well in the target recognition of 2-D image datasets, because it is capable of extracting high-dimensional features for matching with labels [27]. Many CNN variants have been invented to adapt to real-world applications [28]. For example, transfer learning has been introduced into CNN to enhance the flexibility of the network [29], and multilayer autoencoders (AEs) have been employed to improve the robustness of conventional target recognition [30].

Though CNN and its variants have been successfully applied in various scenarios, it still attract great interests in the SAR imaging fields. For accurate recognition, a training dataset with a substantial number of samples and precise labeled data needs to be constructed, which ensures the adequacy and accuracy of feature representation. However, the practical SAR image data are penurious, as data acquisition can be time and cost consuming. In addition, due to the phase error caused by the nonideal and nonsystematic movements and motion errors, the intended SAR image is defocusing. Consequently, it is difficult to obtain a fully focused SAR image as accurate labels to meet the data-driven CNN. Especially, in lightweight airborne platforms, unavoidable motion deviations out of the measurement of onboard navigation unit will defocus the target image. In other words, it would be meaningful to utilize the SAR data even it is defocused due to the nonsystematic phase error, since the fully focused SAR images are difficult to obtain and useful information about the backscattered target has been also involved in the echoed data of the defocused SAR image. Therefore, the defocused SAR images as the inputs for the autofocusing network are of practical significance.

Moreover, due to the limited mathematics in evaluating CNN, there are still no analytical formulations to access the extracted features. Therefore, the features in CNN are random and uninterpretable in specific mathematics. Even when provided with a dataset with sufficient samples, the network still suffer from redundancy in characterizing the features of interest. Because of the poor interpretation in mathematics, it is burdensome to evaluate several typical features that are key components for the target recognition. Consequently, it is valuable to enhance the mathematical interpretation for deep learning networks, which would be useful to improve the accuracy of target recognition under defocused and noised data.

D. Multifeature Optimization

To extract the features of interest of a target, either a model-driven or data-driven approach can be utilized. Both methods have their own merits, where the former builds specific algorithmic models to achieve strong applicability, but the modeling complexity increases dramatically when solving complicated problems, while the latter lacks interpretability. The two approaches combined with each other have great potential. It would be capable of solving complicated inverse problems and possessing meaningful mathematical solutions simultaneously. To incorporate the analytical prior knowledge, feature fusion networks [31] are introduced. To enhance the scattering features in the SAR images, scattering centers are exploited for the accurate SAR recognition under various conditions [32]. Then, for structural features, a hybrid network with structural constraints is designed [33]. It can be observed from [33] that an appropriate feature regularization is introduced in the network architecture. Followed by the development of hybrid model-driven and data-driven networks, an interested feature can be formulated, and the neural network deep learning can be improved with somewhat meaningful intentions. However, in practice, there are several key features that are important for target recognition. The target recognition is affected by the low signal-to-noise ratio caused by the

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additive noises and defocused image responses caused by the multiplicative phase errors. On the one hand, sparse features can be quantified to enhance the dominant scattering of the SAR image to suppress the background noises [34]. On the other hand, the focusing feature can be quantified to ensure the contrast of SAR images to refocus the SAR image. Both sparse feature and focusing feature play a crucial role in improving the target recognition accuracy. Therefore, it is valuable to quantify both the sparse and focusing features in the recognition network. Unfortunately, it is difficult to evaluate both the features in a cooperative way. Without proper coordination of feature quantization, accurate improvement cannot be guaranteed.

In this article, a deep merged autofocusing network (MAFNet) is proposed to simultaneously refocus the SAR imagery and obtain an accurate target recognition result to accommodate to the training dataset in the defocused and noised SAR data. The image entropy (IE) and ℓ_1 norm are employed to evaluate the focusing and sparse features, respectively. The IE is similar to the information entropy in mathematical definition with only a difference of a minus [13]. It is worth noting that the concept of entropy is statistically based, and it is essentially a measure of the uncertainty of a probability distribution. The physical meaning of the IE can be regarded as a metric that is capable of evaluating the uniformity of the distribution of scatters of target of interest. The more uniformity of the distribution, the higher the IE, and the poorer the focusing quality. On the contrary, to achieve a focused image, a smaller IE is the goal for the optimization. In addition, the minimum entropy cannot be simply concluded as representing the image contrast. The physical meaning of the minimum entropy is to achieve a highly concentrated energy distribution of scatters in an image. In such a case, considering the phase error results in the image defocusing, an objective function of IE with respect to the phase error variable can be established. By optimizing or equivalently minimizing the IE function, the phase error can be estimated. Compared with other metrics, such as root mean square error (RMSE), the IE do not need a reference function, which enhances the practicality of the intended algorithm. By incorporating the key features in a novel multivariate loss function, focusing and sparse feature are quantized in a "decomposition–coordination" manner with the aid of a deep unfolding network for the intended target recognition. Because of the unfolding scheme, the intended network properly balances the two features for the target recognition. In order to overcome the nonconvexity of the IE minimization problem, a tight surrogate function is designed to ensure a closed-form solution for the feature. Due to the two key features being quantized, the network is constrained, and the feature extraction becomes purposeful. It is helpful in improving the mathematical interpretation of the network. Furthermore, to accommodate to the invariant data dimensions, the MAFNet is established in a U-shaped architecture. Cross convolution is applied to preserve the continuity of the phase history in the range-compressed data. As a result, even when the training dataset is contaminated with multiplicative phase errors, MAFNet is capable of guaranteeing the accuracy for the SAR recognition. The MSTAR dataset is the most commonly used benchmark dataset in the field of SAR target recognition [35], [36]. Therefore, the canonical single polarization MSTAR dataset is exploited to validate the effectiveness of the proposed MAFNet. Comparisons with conventional algorithms are performed to show the superiority in terms of both quantitative and qualitative aspects.

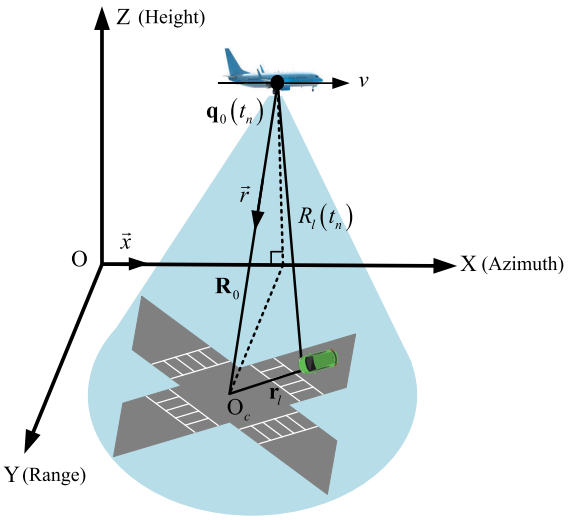


Fig. 1. Generalized airborne SAR geometry.

E. Mathematical Notation

To facilitate the mathematical derivations in this article, the matrix variables are given in bold capital letters. For example, the 2-D range-compressed data are $\mathbf{Y}$. The vector variables are given in bold small letters; for example, the position vector of the lth scatter is $\mathbf{r}_l$. The scalar variables are given in common small letters; for example, the slow-time variable is t_n in Section II.

II. SAR Geometry and Signal Model

Fig. 1 depicts a general airborne SAR geometry, where XYZ-O is the Cartesian coordinates. The airborne SAR is assumed to fly in a straight line in the azimuthal direction at a constant speed v and altitude h. $\mathbf{q}_0(t_n)$ is the position vector of the SAR antenna phase center (APC), and t_n is the slow-time variable. In the ground scene, the observing scene center is given as O_c, and the displacement vector of the lth scatter is $\mathbf{r}_l$. The nearest slant-range vector from the flight track to the scene center is $\mathbf{R}_0$, and the distance from the airborne platform to the scatter on the ground is defined to be $R_l(t_n)$.

Define the orthogonal unit vectors in the azimuth and slant-range directions as $\vec{\mathbf{x}}$ and $\vec{\mathbf{r}}$, respectively. We have $\mathbf{R}_0 = R_0\vec{\mathbf{r}}$ for nearest range vector, and $\mathbf{q}_0(t_n) = u_n\vec{\mathbf{x}}$ can be given as the motion synthesis vector of the airborne platform, where $u_n = vt_n$. When SAR flies along a predetermined track, the predetermined range history is called the systematic movement. Polar format algorithm (PFA) [37] or range migration algorithm (RMA) [38] can be applied to correct the systematic movement. The slant-range between the airborne radar and the lth scatter can be expressed in the Taylor expansion with

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respect to $|\mathbf{R}_0 - \mathbf{q}_0(t_n)|$. The second-order and high orders can be ignored under the far-field assumption [39], and the Taylor expansion can be expressed as follows:

$$\begin{aligned} R_l(t_n) &= |\mathbf{R}_0 + \mathbf{r}_l - \mathbf{q}_0(t_n)| \\ &= |\mathbf{R}_0 - \mathbf{q}_0(t_n)| + \frac{\mathbf{R}_0 - \mathbf{q}_0(t_n)}{|\mathbf{R}_0 - \mathbf{q}_0(t_n)|}\mathbf{r}_l + O(\mathbf{r}_l) \\ &= \sqrt{R_0^2 + (vt_n)^2} + \frac{R_0 r_l^r - (vt_n) r_l^x}{\sqrt{R_0^2 + (vt_n)^2}} + O(\mathbf{r}_l) \end{aligned} \tag{1}$$

where $\mathbf{r}_l = r_l^x \vec{\mathbf{x}} + r_l^r \vec{\mathbf{r}}$ is the position vector of the lth scatter and $O(\mathbf{r}_l)$ is the higher orders that can be ignored under far-field assumption and narrow-beam assumption [40]. Since the first-order expansion in (1) can be further expressed as the dot product of the unit vectors $\vec{\mathbf{x}}$ and $\vec{\mathbf{r}}$, and the dot product of the orthogonal cross term equals 0, the result is a scalar as $R_l(t_n)$ shown in (1).

According to the principle of SAR imagery, the image is generated from the wideband pulses and relative movements between the airborne radar and the target of interest. However, due to atmospheric turbulence, the airborne SAR may fly with deviations from the predetermined trajectory. In such cases, there will be nonideal and nonsystematic motion errors, which will lead to both envelope displacements and phase errors in the echoes. Generally, envelop displacements can be accurately compensated using the airborne INU and GPS. However, the accuracy of correcting phase errors is limited because the azimuth phase is relative to the wavelength, whereas the envelope is relative to the range resolution cell. Due to the uncompensated phase errors, the SAR imagery will be defocused and smeared. Accordingly, it is difficult to ensure an accurate target recognition.

Define $\phi(k, t_n) = k\Delta r(t_n)$ to be the phase modulation due to the nonideal and nonsystematic motion errors $\Delta r(t_n)$, where k is the transmitting wavenumber variable or the spatial frequency of the transmitted pulses. In practice, the total range history includes the systematic and nonsystematic range functions. After range deramping of transmitting signal modulation [41], the echoes of the reference channel in (1) can be explicitly written as follows:

$$\begin{aligned} &S(k, t_n) \\ &= \sum_{l=1}^{L} \sigma_l \exp\left[-\mathrm{j}k\left(\frac{R_0 r_l^r + u_n r_l^x}{\sqrt{R_0^2 + u_n^2}}\right)\right] \exp\left[\mathrm{j}\phi(k, t_n)\right] + N(k, t_n) \end{aligned} \tag{2}$$

where $N(k, t_n)$ is the additive noise and k can be specifically written as $k = 4\pi f/c$, where $f = f_0 + \Delta f$ is the transmitting frequency, f_0 is the centroid frequency, Δf is the frequency variation, and c is the speed of light. Commonly, considering the nonsystematic phase error [14] and applying the PFA [37] to eliminate the systematic movement in (2), the echoes in (2) can be expressed as follows:

$$\begin{aligned} &S(k_r, k_x) \\ &= \sum_{l=1}^{L} \sigma_l \exp\left[-\mathrm{j}\left(k_r r_l^r + k_x r_l^x\right)\right] \exp\left[\mathrm{j}\phi\left(\tilde{t}_n\right)\right] + N(k_r, k_x) \end{aligned} \tag{3}$$

where $k_r = kR_0/(R_0^2 + u_n^2)^{1/2}$ and $k_x = k_r u_n / R_0$ denote the range and azimuth wavenumber in the post-PFA data domain. Two subsequent interpolators can be applied for the polar format interpolation from (k, t_n) to (k_r, k_x) [42]. After PFA, $k_x = (-vk_0/R_0)\tilde{t}_n$ can be expressed with a linear relation with a new slow-time variable $\tilde{t}_n$. It indicates that the azimuth wavenumber k_x is inherently an interpolated azimuth slow-time $\tilde{t}_n$. In such a case, the azimuth phase error (APE) function can be written as $\phi(\tilde{t}_n)$.

By applying the Fourier transformation to k_r and replacing k_x with $\tilde{t}_n$ for the signal in the range-compressed domain $(\hat{r}, \tilde{t}_n)$, the range-compressed form of (3) can be given as follows:

$$\begin{aligned} S\left(\hat{r}, \tilde{t}_n\right) = &\sum_{l=1}^{L} \sigma_l \mathrm{sinc}\left(\hat{r} - r_l^r\right) \exp\left(\mathrm{j}2\pi \frac{2v r_l^x}{\lambda R_0} \tilde{t}_n\right) \exp\left[\mathrm{j}\phi\left(\tilde{t}_n\right)\right] \\ &+ N\left(\hat{r}, \tilde{t}_n\right) \end{aligned} \tag{4}$$

where $\hat{r}$ denotes the slant-range index, k_x is substituted by $\tilde{t}_n$ to express the signal in the range-compressed domain $(\hat{r}, \tilde{t}_n)$, the phase error can be written as $\phi(\tilde{t}_n) = k_0 \Delta r(\tilde{t}_n)$, and $k_0 = 4\pi/\lambda$ gives the wavenumber centroid. In (4), $\mathrm{sinc}(\hat{r} - r_l^r)$ is the range envelop function of ground scatters. The sinc function directly denotes the slant-range position of the scatter and $N(\hat{r}, \tilde{t}_n)$ is the additive noise.

To express the signal model in a concise manner, (4) can be written in the matrix form as follows:

$$\mathbf{Y} = \mathbf{EAX} + \mathbf{C}_\mathrm{N} \tag{5}$$

where $\mathbf{Y} \in \mathbb{C}^{N \times M}$ denotes the range-compressed echoes as $S(\hat{r}, \tilde{t}_n)$ in (4), N represents the number of pulses, and M corresponds to the number of range bins. $\mathbf{X} \in \mathbb{C}^{N \times M}$ in (5) is the intended SAR image or features of interests, $\mathbf{A}$ is the Fourier dictionary with each element established as $\mathbf{A} = [\mathbf{a}(\hat{f}_d^1), \ldots, \mathbf{a}(\hat{f}_d^N)]$ and $\mathbf{a}(\hat{f}_d^n) = [e^{-\mathrm{j}2\pi \hat{f}_d^n \tilde{t}_1}, \ldots, e^{-\mathrm{j}2\pi \hat{f}_d^n \tilde{t}_N}]^\mathrm{T}$, where $\hat{f}_d^n$ indicates the Doppler frequency index and $[\cdot]^\mathrm{T}$ denotes the transpose operator. In (5), based on the azimuth invariant assumption, the dictionary $\mathbf{E}$ can be formed as a diagonal matrix with the diagonal elements for the time-variant phase errors. It represents the APE $\phi(\cdot)$. $\mathbf{C}_\mathrm{N}$ denotes the noise matrix. In this article, (5) is used to formulate the signal model, where $\mathbf{E}$ is the multiplicative phase error and $\mathbf{C}_\mathrm{N}$ is the additive noise error. The SAR imagery is to solve the inverse problem in (5), and closed-form solutions can be obtained for the features of interest. However, due to multiplicative phase errors and additive noises, the intended target recognition will become poor, since the target image will be defocused and with low signal-to-noise ratio. Therefore, to refocus the SAR image and obtain the high target recognition accuracy, the generalization ability to adapt the additive noise error, and multiplicative phase error is primarily concerned in this article.

III. Preliminaries

Inherently, the SAR imagery is to solve the inverse problem, as shown in (5). Simply, when the multiplicative error $\mathbf{E}$ and the additive error $\mathbf{C}_\mathrm{N}$ are not taken into account, it can be solved as $\mathbf{X} = \mathbf{A}^{-1}\mathbf{Y}$. However, in practice, due to the motion errors and unavoidable noises, the inverse problem is ill-posed. The image quality will be degraded, where the

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focusing performance will be poor and the effective SNR will be low. It will definitely degrade the performance of further target recognition.

A. Autofocusing of Nonsystematic Errors

Due to the nonideal and unknown movement of the airborne platform from atmospheric turbulence and navigational errors, the nonsystematic motion error occurs. Accordingly, the SAR image will be defocused, because of the phase error caused by the nonideal and nonsystematic motion errors. To refocus the SAR image, the MEA algorithm would be optimal in strong noise scenario [43]. Moreover, it has no limitation on the form of errors and the dominant scatters. Thus, the MEA algorithm is used in this article to solve ill-posed of the inverse problem mentioned in (5). The phase error caused by the nonsystematic motion errors is defined to be $\phi(\tilde{t}_n)$, the objective function about the phase error for the minimum IE can be expressed in the constrained problem as follows:

$$\begin{cases} \hat{\phi} = \arg\ \min \|\mathbf{X}(\phi)\|_{\mathrm{EN}} \\ \|\mathbf{X}(\phi)\|_{\mathrm{EN}} = \dfrac{-1}{E_X} \displaystyle\sum_{m,n} \|\mathbf{X}(\phi)\|_{\mathrm{F}}^2 \cdot \ln \|\mathbf{X}(\phi)\|_{\mathrm{F}}^2 + \ln E_X \end{cases} \tag{6}$$

where $\|\cdot\|_{\mathrm{EN}}$ stands for the entropy norm regularizer and $\|\cdot\|_{\mathrm{F}}$ stands for the Frobenius norm. In (6), $\hat{\phi}$ represents the nonsystematic phase error to be estimated and $\mathbf{X}(\phi)$ represents the function with respect to the phase error ϕ. Specifically, it can be written as $\mathbf{X}(\phi) = (1/N) \sum_{i=0}^{N-1} \exp(\mathrm{j}2\pi(ni/N))\exp(\mathrm{j}\phi_i)\mathbf{Y}$, where $\mathbf{Y}$ is the range compressed data as formulated in (5). The phase error resulting from the nonsystematic motion error will contaminate the range-compressed data, and the image result can be expressed as the Fourier transformation of the range-compressed data. Furthermore, in (6), $\sum_{m,n} \|\mathbf{X}(\phi)\|_{\mathrm{F}}^2$ represents the total energy of the SAR image.

Next, to achieve an accurate estimation of ϕ, it is required to solve the optimization problem in (6). However, due to the logarithmic operator in (6), the optimization is not a convex problem, and there is no closed-form solution for the error function $\phi(\tilde{t}_n)$. It would be difficult to obtain an accurate and focused SAR image for target recognition.

B. Model-Driven Feature Quantization

When the problem in (5) is under-determinated or error-contaminated, the solution will be ill-posed. To this end, a priori knowledge can be incorporated to relax the solution. In general, a proper norm function can be introduced into the optimization in a regularization manner. Regarding to the SAR imagery, when a specific image feature is to be quantized or enhanced, based on the augmented Lagrange function, a constrained optimization problem such as (6) can be transformed into an unconstrained optimization problem as follows:

$$\mathbf{X}_{\mathrm{re}} = \arg \min_{\mathbf{X}} \|\mathbf{X}\|_{\mathrm{norm}} + \frac{1}{2}\|\mathbf{Y} - \mathbf{E}\mathbf{A}\mathbf{X}\|_{\mathrm{F}}^2 \tag{7}$$

where $\mathbf{X}_{\mathrm{re}}$ denotes the recovered SAR image/feature and $\|\mathbf{X}\|_{\mathrm{norm}}$ is the generalized regularizer to represent specific priors. The second term in (7) is the fidelity term measuring the recovering accuracy.

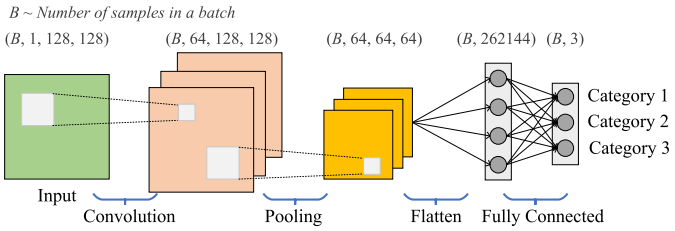


Fig. 2. Architecture of conventional CNN.

Conventionally, when a sparse feature is to be quantized, $\|\mathbf{X}\|_1$ can be formulated as the regularizer, and the regularization problem becomes the classic least absolute shrinkage and selection operator (LASSO) problem [44]. Due to the ℓ_1 norm function is nondifferentiable, it is difficult to achieve a closed-form solution for the sparse feature. To this end, the proximal algorithm [10], [45] is employed to derive a soft-thresholding operator for the feature. Furthermore, to quantize structural features of the target of interest, $\|\mathbf{X}\|_{\mathrm{TV}}$ can be used for an ℓ_{TV} norm regularizer. In such a case, structure-driven SAR imagery becomes popular [46]. However, when complicated regularizers are to be incorporated, the objective function may be nonconvex. It is still hard to guarantee closed-form solutions. Therefore, it is necessary to design a tight convex surrogate function to ensure the intended features to be quantized in a closed-form solution.

C. Data-Driven SAR Recognition

In order to extract features for target recognition, the data-driven method can be utilized. It is capable of learning target features that are essential for representing different categories from a large scale of samples. The most classical method for target recognition is a CNN based on supervised learning. Conventional CNN is data-driven. It highly depends on a large amount of high-quality training data and high-precision labeled data. Owing to the strong ability of nonlinear fitting, CNN is capable of accurately capturing the mapping relationship between images and categories from a large number of samples. In such cases, CNN has superior recognition accuracy beyond other methods. Fig. 2 shows a schematic of triple classification using the conventional CNN.

In Fig. 2, assuming that the input SAR imagery is the size of 128 * 128 resolution cells, the dimension of the input blocks is (B, 1, 128, 128). Then, two convolutional layers with a kernel size of 3 and output channel size of 64 are utilized for feature extraction, and the dimension of the feature map becomes (B, 64, 128, 128). Max-pooling layer of 2 * 2 is employed to discard redundant features, and the output feature map is in dimension of (B, 64, 64, 64). Moreover, a feature map with dimension of (B, 64 * 64 * 64) = (B, 262 144) is acquired by flattening, and a batch of triple classification predicted categories can be obtained via fully connected layers.

However, the CNN-based networks have some problems in practices. On the one hand, it is difficult for the conventional CNN to accommodate to the complex-valued training data. Specifically, the complex-valued SAR data are inherently a phase history data, where the phase of the data is continuous and related to the range variation between the radar and the target. Therefore, it will be meaningful

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to develop a novel network that is capable of utilizing to the complex-valued SAR data. On the other hand, the data-driven CNN-based networks lack quantizable constraints for feature extraction. Due to the data-driven CNN lacks a mathematical model of the features, two key focusing and sparse features for target recognition that affect the SAR target recognition accuracy cannot be quantified. In such cases, the extracted features are random and redundant. As a result, the interpretability of the extracted feature is poor, and there is no resolvable expression for the extracted features. The performance of the network can only depend on a large amount of high-quality training data. When the train data amount is insufficient or the data quality is poor, the accuracy of target recognition will be decreased dramatically.

IV. Deep MAFNet for SAR Recognition

Considering that the focusing quality of the training data may be poor, we propose a deep MAFNet, which is capable of quantitatively extracting key features for the SAR recognition. The framework of the MAFNet is an end-to-end network, including a focusing module and a recognition module. The two modules are embedded into a single framework. The focusing module and recognition module are designed for refocusing the SAR image and target recognition, respectively.

In the focusing module, the IE and ℓ_1 norm are used as prior for the algorithmic models are first designed to quantitatively evaluate the focusing and sparse features, respectively. Then, a U-shaped subnetwork is constructed in a model-driven manner to adapt to the complex-valued SAR data. More specifically, the model-driven quantization is developed in a cooperative manner, where the algorithmic models are optimized and solved in closed-form solutions, and corresponding dual variables are introduced to measure the residual errors for a global optimization. Moreover, a deep unfolding technique is utilized in the feature quantization process to enable the regularization coefficients to be learnable to improve the interpretability of the network. Finally, the recognition module is implemented to predict the categories.

A. Focusing Module

In order to improve the focusing quality of SAR data, we collaboratively quantize two key features in the SAR data, including the focusing feature and the sparse feature.

1) Deep Multifeature Quantization: In general, the scattering characteristics of the target of interest are important for target recognition. Specifically, focusing and sparse features are two typical scattering features, where the former generally ensures the outline of the target, while the latter represents the dominant scatters of the target [10]. Therefore, for SAR recognition, the focusing and sparse features would be the most important. It would be feasible to make the data-driven U-shaped subnetwork to extract the specific features for an accurate recognition even in a penurious and defocused dataset.

In this work, both focusing and sparse features are to be evaluated. The priors are to be formulated as the IE norm regularization for the focusing feature and the ℓ_1 norm regularization for the sparse feature. Accordingly, the problem can be formulated into a joint optimization objective function of the focusing feature and sparse feature as follows:

$$\tilde{\mathbf{X}} = \arg\min_{\mathbf{X}} \frac{1}{2}\|\mathbf{Y}_{\mathrm{L}} - \mathbf{E}\mathbf{A}\mathbf{X}\|_2^2 + \lambda_1\|\mathbf{X}\|_{\mathrm{EN}} + \lambda_2\|\mathbf{X}\|_1 \tag{8}$$

where $\mathbf{Y}_{\mathrm{L}}$ denotes range-compressed data under a defocused image that is to be evaluated and $\|\cdot\|_1$ denotes the ℓ_1 norm. It can be noted that the physical meaning of IE is to measure the uniformity of the distribution of scatters distribution in an image. The lower the entropy, the more concentrated the distribution of scatters, and the better focused the image.

In this article, considering that conventional methods for multiregularization optimization require complicated high-dimensional matrix operations, a "decomposition–coordination" scheme is designed to collaboratively quantize the focusing feature and the sparse feature of SAR images. In such a case, the joint multifeature quantization can be decomposed into multiple subproblems of local feature quantization, facilitating collaborative execution. In (8), two priors are regularized, which is a typical joint optimization of multitask learning. In order to establish the "decomposition–coordination" framework, two splitted variables $\mathbf{Z}_1$ and $\mathbf{Z}_2$ are introduced. Then, the relationship between the primal variable $\mathbf{X}$ and splitted variable is applied as a constraint, and the problem in (8) can be equivalently transformed into [47]

$$\arg\min\{f(\mathbf{X}) + e(\mathbf{Z})\} \ \text{s.t.} \ \mathbf{P}\mathbf{X} + \mathbf{Q}\mathbf{Z} = \mathbf{0} \tag{9}$$

where $\mathbf{Z} = [\mathbf{Z}_1, \mathbf{Z}_2]$, $f(\mathbf{X}) = (1/2)\|\mathbf{Y}_{\mathrm{L}} - \mathbf{E}\mathbf{A}\mathbf{X}\|_2^2$ denotes the fidelity term and $e(\mathbf{Z})$ is the combination of the regularizers. In $e(\mathbf{Z})$, the sparse and focusing features are considered, and it is the combination of ℓ_1 norm and entropy norm, as shown in (8). Setting $\mathbf{P} = [\mathbf{I}, \mathbf{I}]^{\mathrm{T}}$, where $\mathbf{I}$ is the unit matrix, and $\mathbf{Q} = -\mathbf{I}$, the two split variables can be explicitly shown as $\mathbf{Z}_1 = \mathbf{Z}_2 = \mathbf{X} \in \mathbb{C}^{N\times M}$. Next, by incorporating a dual variable $\mathbf{U}$, an augmented Lagrangian function of (9) can be derived as follows:

$$\begin{aligned} &L_\rho(\mathbf{X}, \mathbf{Z}; \mathbf{U}) \\ &\quad = f(\mathbf{X}) + g(\mathbf{Z}) + \mathbf{U}^{\mathrm{H}}(\mathbf{X} - \mathbf{Z}) + \frac{\rho}{2}\|\mathbf{X} - \mathbf{Z}\|_{\mathrm{F}}^2 \end{aligned} \tag{10}$$

where $[\cdot]^{\mathrm{H}}$ denotes the conjugate transpose operator, $\mathbf{U}$ is the dual variable representing the residual error between the primal variable $\mathbf{X}$ and the split variable $\mathbf{Z}$, and ρ is the penalty coefficient.

The variables in the augmented Lagrangian problem (10) can be solved by gradient descent [47]. When one variable is to be solved, the rest of the variables should be fixed. The iteration of all the variables in (10) can be achieved via the "decomposition–coordination" procedure [47] as follows:

$$\begin{cases} \text{for} \ \ g = 1:2 \\ \mathbf{Z}_g^{k+1} = \arg\min_{\mathbf{Z}_g}\left\{\lambda_g \cdot e\left(\mathbf{Z}_g\right) + \dfrac{\rho}{2}\left\|\mathbf{X}^k - \mathbf{Z}_g + \mathbf{D}_g^k\right\|_{\mathrm{F}}^2\right\} \\ \mathbf{D}_g^{k+1} = \mathbf{D}_g^k + \mathbf{X}^k - \mathbf{Z}_g^{k+1} \\ \text{end} \\ \mathbf{Z}^{k+1} = \left[\mathbf{Z}_1^{k+1}, \mathbf{Z}_2^{k+1}\right] \\ \mathbf{D}^{k+1} = \left[\mathbf{D}_1^{k+1}, \mathbf{D}_2^{k+1}\right] \\ \mathbf{X}^{k+1} = \arg\min_{\mathbf{X}}\left\{f(\mathbf{X}) + \dfrac{\rho}{2}\left\|\mathbf{X} - \mathbf{Z}^{k+1} + \mathbf{D}^{k+1}\right\|_{\mathrm{F}}^2\right\} \end{cases} \tag{11}$$

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where the superscript k denotes the iteration number. In (11), the "for-end" loop represents the optimization for the feature variables $\mathbf{Z}_1$ and $\mathbf{Z}_2$. The "decomposition–coordination" procedure can be divided into three steps. In the first step, the variables $\mathbf{Z}_1^{k+1}$ and $\mathbf{D}_1^{k+1}$ of the focusing feature are updated. In the second step, the variables $\mathbf{Z}_2^{k+1}$ and $\mathbf{D}_2^{k+1}$ of the sparse feature are updated. In the third step, the variables calculated above are gathered to update the coordinated variables $\mathbf{Z}^{k+1}$, $\mathbf{D}^{k+1}$, and finally update $\mathbf{X}^{k+1}$. According to the coordinate descent solution for the augmented Lagrangian function, the joint optimization for (10) is decomposed into two local optimizations for $\mathbf{Z}_1$ and $\mathbf{Z}_2$. Simultaneously, the scaled dual variable $\mathbf{D} = \mathbf{U}/\rho$ is updated to measure the residual errors between the related feature variable and the primal variable. After each feature variable and dual variable are updated, they are collected into $\mathbf{Z}$ and $\mathbf{D}$. Under the updated $\mathbf{Z}^{k+1}$ and $\mathbf{D}^{k+1}$, finally, the image variable $\mathbf{X}$ is updated, where both the updated feature variables and residual errors are coordinated. In (11), it is obvious that the framework executes in a "decomposition–coordination" manner, which facilitates the solution for complicated joint problem, and simultaneously guarantees the accuracy of the global solution by coordinating both the feature variables and the corresponding residual errors.

Detailed mathematics for the feature quantization in the "decomposition–coordination" framework are given as follows.

Focusing Feature Quantization: According to (11), for updating the focusing feature variable $\mathbf{Z}_1$, the optimization function can be written in proximal optimization as follows:

$$\mathbf{Z}_1^{k+1} = \arg\min_{\mathbf{Z}_1}\left\{\lambda_1\|\mathbf{Z}_1\|_{\mathrm{EN}} + \frac{\rho}{2}\left\|\mathbf{X}^k - \mathbf{Z}_1 + \mathbf{D}_1^k\right\|_{\mathrm{F}}^2\right\} \tag{12}$$

where λ_1 stands for the regularization parameter of the entropy norm and $\mathbf{D}_1$ is the residual error. Considering the entropy norm regularizer is a function of the azimuthal phase error, the optimization of $\mathbf{Z}_1$ is expressed as a function of the azimuthal phase error $\phi(\tilde{t}_n)$ as follows:

$$\mathbf{Z}_1 = \frac{1}{N}\sum_{i=0}^{N-1}\exp(\mathrm{j}2\pi(ni/N))\exp(\mathrm{j}\phi_i)\mathbf{Y}_{\mathrm{L}} \tag{13}$$

where $\phi_i = \phi(\tilde{t}_i)$ denotes the ith moment of the phase error.

Since $g(p) = -p\ln p$ is a real-valued concave function for $p > 0$, the IE in (6) can be rewritten in the form of a concave function with respect to the phase error ϕ as $\|\mathbf{X}(\phi)\|_{\mathrm{EN}} = (1/E_X)\sum_{m,n} g(\|\mathbf{X}(\phi)\|_{\mathrm{F}}^2) + \ln E_X$, which implies that there is no closed-form solution for the minimization of the IE regularization optimization in (12). To solve such a nonconvex problem, it needs to be transformed into a surrogate function that is an upper bound on the objective at the kth iteration. In general, the surrogate function should be the tightest one, which satisfies

$$\begin{cases}\Omega\left(\mathbf{Z}_1^k;\mathbf{Z}_1^k\right) = \Psi(\mathbf{Z}_1)|_{\mathbf{Z}_1=\mathbf{Z}_1^k}\\ \Omega'\left(\mathbf{Z}_1;\mathbf{Z}_1^k\right)\big|_{\mathbf{Z}_1=\mathbf{Z}_1^k} = \Psi'(\mathbf{Z}_1)\big|_{\mathbf{Z}_1=\mathbf{Z}_1^k}\\ \Omega\left(\mathbf{Z}_1;\mathbf{Z}_1^k\right) \geq \Psi(\mathbf{Z}_1)\end{cases} \tag{14}$$

where $\Psi(\mathbf{Z}_1)$ stands for the original cost function, while $\Omega(\mathbf{Z}_1, \mathbf{Z}_1^k)$ is the intended surrogate function when setting some of input as $\mathbf{Z}_1^k$, and $\Omega'(\cdot)$ and $\Psi'(\cdot)$ denote the first-order derivatives. From (14), first, the surrogate function should be equal to the original function when setting $\mathbf{Z}_1 = \mathbf{Z}_1^k$. Second, the derivatives of the original and surrogate functions are equal to each other at the same point. Finally, the surrogate should be above on the original one. Accordingly, the surrogate objective function is given [10] as follows:

$$\begin{aligned}&\Omega\left(\mathbf{Z}_1;\mathbf{Z}_1^k\right)\\ &= -\frac{\lambda_1}{E_Z}\sum_{m,n}\|\mathbf{Z}_1(n,m)\|_{\mathrm{F}}^2\ln\left\|\mathbf{Z}_1^k(n,m)\right\|_{\mathrm{F}}^2 + \lambda_1\ln E_Z\\ &\quad + \left(\frac{\rho}{2} - \frac{\lambda_1}{E_Z}\right)\sum_{m,n}\|\mathbf{Z}_1(n,m)\|_{\mathrm{F}}^2 + \frac{\lambda_1}{E_Z}\sum_{m,n}\left\|\mathbf{Z}_1^k(n,m)\right\|_{\mathrm{F}}^2\\ &\quad + \frac{\rho}{2}\left\|\mathbf{X}^k + \mathbf{D}_1^k\right\|_{\mathrm{F}}^2\end{aligned} \tag{15}$$

where the logarithm term in the IE norm is fixed with the latest value of $\mathbf{Z}_1$. In (15), it is the designed cost function based on the criterion in (14). It can be observed from (15) that $\Omega'(Z(\phi_i); Z(\phi_i^k)) = -\Omega'''(Z(\phi_i); Z(\phi_i^k))$, where the first-order derivative equals to the negative of the third-order derivative. In such a case, the intended focusing feature $\mathbf{Z}_1$ or the phase error $\phi(\tilde{t}_n)$ can be easily solved under the high-order derivative function. Accordingly, each element of the phase error function $\phi(\tilde{t}_n)$ can be updated as follows:

$$\phi_i^{k+1} = \phi_i^k + \arctan\left(\frac{\vartheta_i}{\xi_i}\right) \tag{16}$$

where $\vartheta_i = \Omega_i'|_{\phi_i=\phi_i^k}$ and $\xi_i = -\Omega_i''|_{\phi_i=\phi_i^k}$. In (16), it is the solution to the phase error calculated on the cost function in (15). When the elements of the phase error are updated, the diagonal matrix can be obtained as $\mathbf{E}^{k+1}$. As a result, the solution of phase error compensation can be expressed as follows:

$$\mathbf{Z}_1^{k+1} = \mathbf{A}^{\mathrm{H}}\left(\mathbf{E}^{k+1}\right)^{\mathrm{H}}\mathbf{Y}_{\mathrm{L}}. \tag{17}$$

The dual variable $\mathbf{D}_1$ can be updated as follows:

$$\mathbf{D}_1^{k+1} = \mathbf{D}_1^k + \mathbf{X}^k - \mathbf{Z}_1^{k+1} \tag{18}$$

where dual variable measures the residual error between primal variable $\mathbf{X}^k$ and feature variable $\mathbf{Z}_1^{k+1}$.

Sparse Feature Quantization: Next, according to (11), the sparse feature variable $\mathbf{Z}_2$ is to be updated as follows:

$$\mathbf{Z}_2^{k+1} = \arg\min_{\mathbf{Z}_2}\left\{\lambda_2\|\mathrm{vec}(\mathbf{Z}_2)\|_1 + \frac{\rho}{2}\left\|\mathbf{X}^k - \mathbf{Z}_2 + \mathbf{D}_2^k\right\|_{\mathrm{F}}^2\right\} \tag{19}$$

where λ_2 stands for the regularization parameter of the ℓ_1 norm and $\mathrm{vec}(\cdot)$ stands for the vectorization operator of the matrix $\mathbf{Z}_2$. In (19), it is the result of the second step of the "decomposition–coordination," where $\mathbf{Z}_1$ and $\mathbf{X}$ are fixed to calculate the variable $\mathbf{Z}_2$. It can be noted from (19) that the problem is inherently a typical Lasso formulation. Although it is convex, the problem is nondifferentiable. To this end, the proximal algorithm [48] is employed. In accordance, the proximal algorithm of the ℓ_1 norm regularizer is employed to solve the nondifferentiable optimization function in (19), where the proximity operator, namely soft-thresholding operator [49], is utilized to solve the optimization function. Therefore, the

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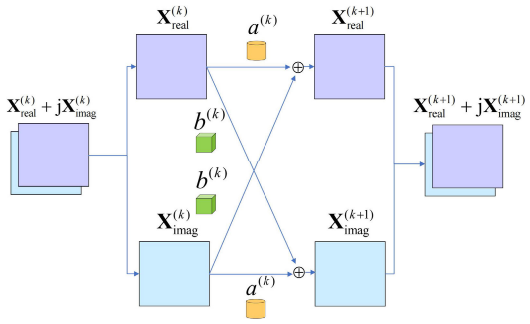


Fig. 3. Diagram of cross convolution.

solution of the sparse feature $\mathbf{Z}_2$ can be expressed as follows:

$$\mathbf{Z}_2{}^{k+1} = \text{Prox}_{\lambda_2/\rho}\left(\mathbf{S}^{k+1}\right) \tag{20}$$

where $\text{Prox}_{\lambda_2/\rho}(\cdot)$ is the soft-thresholding operator, and $\mathbf{S}^{k+1} = \mathbf{A}^{\text{H}}(\mathbf{E}^{k+1})^{\text{H}}\mathbf{A}\mathbf{X}^k + \mathbf{D}_2^k$. The analytical expression of soft-thresholding operator is given as follows:

$$\text{Prox}_{\lambda_2/\rho}\left(\mathbf{S}^{k+1}\right) = \begin{cases} \mathbf{S}^{k+1} - \lambda_2/\rho, & \mathbf{S}^{k+1} > \lambda_2/\rho \\ 0, & \left|\mathbf{S}^{k+1}\right| \le \lambda_2/\rho \\ \mathbf{S}^{k+1} + \lambda_2/\rho, & \mathbf{S}^{k+1} < -\lambda_2/\rho \end{cases} \tag{21}$$

where the gradient of $\text{Prox}_{\lambda_2/\rho}(\mathbf{S}^{k+1})$ with respect to the object $\mathbf{S}^{k+1}$ in (21) can be either 0 or 1. When the value of the object $\mathbf{S}^{k+1}$ is greater than the value of parameter λ_2/ρ, the result of the soft-thresholding operator is equal to $(\mathbf{S}^{k+1} - \lambda_2/\rho)$. When the value of the object $\mathbf{S}^{k+1}$ is less than $-\lambda_2/\rho$, the result of the soft-thresholding operator is equal to $(\mathbf{S}^{k+1} + \lambda_2/\rho)$. Otherwise, the result of the soft-thresholding operator is equal to 0. The solution of the dual variable $\mathbf{D}_2$ is given as follows:

$$\mathbf{D}_2^{k+1} = \mathbf{D}_2^k + \mathbf{X}^k - \mathbf{Z}_2^{k+1}. \tag{22}$$

2) U-Shaped Subnetwork $\mathcal{U}$: After the focusing and sparse features are quantized in a hyperparametric subnetwork, a U-shaped subnetwork is designed to embed the quantizations in the focusing module, where feature extraction is constrained by the focusing loss function. Considering the input SAR data are complex-valued, the architecture of the focusing module network is designed in a U-shaped form, which ensures the dimension of the network output is consistent with the input data [50], [51], [52]. Moreover, compared with the fully convolutional network, the intended U-shaped network is capable of accommodating the associated and continuous phase history data in the range compressed data domain. In such cases, it is feasible to extract the focusing feature. As shown in Fig. 3, the cross convolution is applied for the complex-valued data in the convolutional layers.

The designed cross-convolutional framework accommodates to the complex-valued SAR data by extracting the real and imaginary parts of the input data, respectively. Assuming that the complex-valued kernel of the kth convolutional layer is $a^{(k)} + \text{j}b^{(k)}$ and that of the output is $\mathbf{X}^{(k)} = \mathbf{X}_{\text{real}}^{(k)} + \text{j}\mathbf{X}_{\text{imag}}^{(k)}$, the complex-valued data of the $(k+1)$th convolutional layer can be updated as follows:

$$\begin{aligned} \mathbf{X}_{\text{real}}^{(k+1)} + \text{j}\mathbf{X}_{\text{imag}}^{(k+1)} = &\left[a^{(k)} * \mathbf{X}_{\text{real}}^{(k)} - b^{(k)} * \mathbf{X}_{\text{imag}}^{(k)}\right] \\ &+ \text{j}\left[a^{(k)} * \mathbf{X}_{\text{imag}}^{(k)} + b^{(k)} * \mathbf{X}_{\text{real}}^{(k)}\right]. \end{aligned} \tag{23}$$

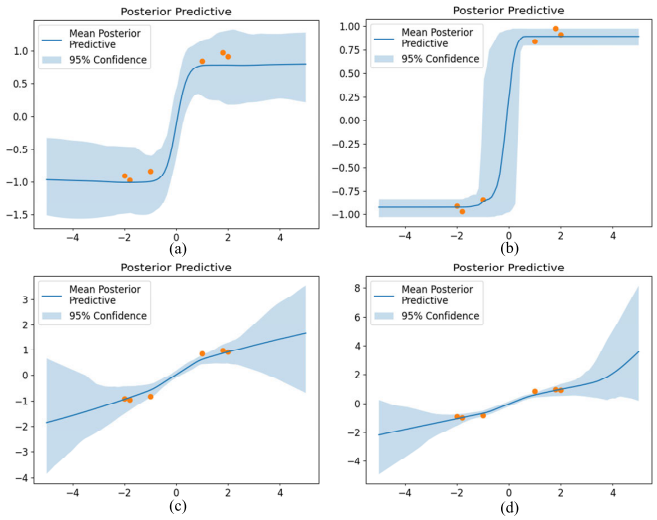


Fig. 4. Predictive posterior with different activation functions. (a) One hidden layer activated by sigmoid. (b) Two hidden layers activated by sigmoid. (c) One hidden layer activated by ReLU. (d) One hidden layer activated by LeakyReLU.

In (23), the imagery part of the cross convolutional output is a combination of the sum of the kernel and the input data, while the real part of the output is a combination of the difference between the kernel and the input data. Finally, the real and imaginary parts of the cross convolution are combined in the space domain via torch. Stack to form the output.

Next, to improve the nonlinearity of the intended network, the LeakyReLU function [53] is adopted as the activation function as follows:

$$y_i = \begin{cases} x_i, & \text{if } x_i \ge 0 \\ \mu_i x_i, & \text{if } x_i < 0 \end{cases} \tag{24}$$

where $\mu_i \in (0, 1)$ stands for a fixed parameter, which is set to be 10^{-2} in this article. In (24), since the function has a small positive slope in the negative region, the problem of neuronal death can be avoided, and the network converges quickly. To intuitively evaluate the effect of LeakyReLU function, a simple posterior prediction experiment is conducted, and the result is shown in Fig. 4. In Fig. 4, the solid line represents the mean of the posterior prediction, and the shaded area stands for the 95% confidence interval area. A 95% confidence interval is a method used in statistics to evaluate the scale of a parameter. In Fig. 4(d), the 95% confidence interval area of the aggregation area of the discrete points is small, while the 95% confidence interval area of the area on both sides of the discrete points is large. In other words, the confidence interval area of the key prediction region is smaller, which means the result of LeakyReLU is appropriate. On the other hand, the area of the two sides with little influence in the prediction is larger, which indicates the stronger generalization ability of LeakyReLU. Therefore, LeakyReLU is selected to be the activation function of the intended MAFNet.

Under the definitions of the cross convolution and the LeakyReLU function in (23) and (24), respectively, the U-shaped subnetwork is established, as shown in Fig. 5. To ensure the dimensional consistency, the architecture of the U-shaped subnetwork is designed in downsampling and

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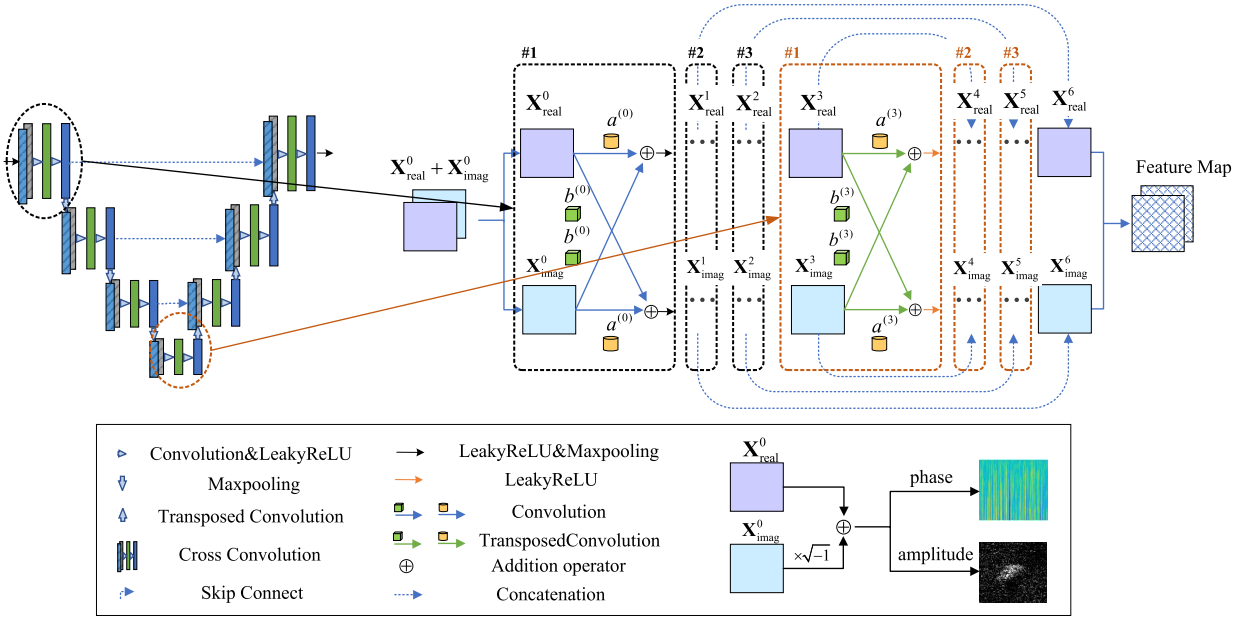


Fig. 5. Diagram of U-shaped subnetwork.

upsampling manners. Specifically, the size of each convolutional kernel is set to be 3×3. The stride size of each cross-convolutional layer is set to be 1, and the padding size is set to be 1. To enlarge the receptive field while reducing the loss of resolution, the dilation rate is set to be 1. To reduce the redundant features, the max-pooling is applied for downsampling, and the transposed cross convolution is utilized for the upsampling.

Furthermore, to accommodate the defocused SAR data, the U-shaped subnetwork should be capable of autofocusing the defocused and noised dataset. According to (5), the resultant nonsystematic phase error can be formulated as a multiplicative diagonal matrix of the intended focused SAR image. In such a case, the defocused SAR data can be formulated as $\mathbf{Y}_{\mathrm{L}} = \mathbf{Y} = \mathbf{EAX} + \mathbf{C}_{\mathrm{N}}$. The input of the network is defined to be $\tilde{\mathbf{Y}}$, and the output of the U-shaped subnetwork can be expressed as $\hat{\mathbf{Y}} = s(\tilde{\mathbf{Y}})$, where $s(\cdot)$ is the response function of the U-shaped subnetwork. The input of the deep multifeature quantization is defined to be $\mathbf{Y}_{\mathrm{L}}$, and the output can be expressed as $\tilde{\mathbf{Y}} = h(\mathbf{Y}_{\mathrm{L}})$, where $h(\cdot)$ is the process of deep multifeature quantization.

It should be noted that it is difficult to achieve focusing module with the data-driven network alone. On the one hand, the loss function is in the form of least squares, and the process of SAR data recovery lacks prior constraints; thus, the process without the guidance of a mathematical model will be prone to over-fitting. On the other hand, there is no specific feature model to guide the solution of the focusing features, and the focusing module has limited generalization ability and strong dependence on the dataset. To this end, the focusing module is developed in a model-driven manner, where the focusing and sparse prior model are simultaneously embedded as the prior constraints within the data-driven U-shaped subnetwork, so that the modeled features can be directly used for target recognition.

Define the SAR data recovered from $\mathbf{Y}_{\mathrm{L}}$ as $\hat{\mathbf{Y}}$, the loss function for autofocusing can be expressed as follows:

$$\mathcal{L}_{\mathrm{F}} = \left\|\hat{\mathbf{Y}} - \mathbf{Y}_{\mathrm{H}}\right\|_{\mathrm{F}}^{2} \tag{25}$$

where $\mathbf{Y}_{\mathrm{H}} = \mathbf{AX}_{\mathrm{H}}$ is the high-quality data in the training dataset. Then, the hyperparameters of the model-driven network are adjusted during the training process guided by the loss function, so that the defocused input data can be refocused.

3) Deep Multifeature Unfolding: After $\mathbf{Z}_1^{k+1}$, $\mathbf{D}_1^{k+1}$ and $\mathbf{Z}_2^{k+1}$, $\mathbf{D}_2^{k+1}$ are collected in (17), (18), (20), and (22), the global optimization for the primal variable $\mathbf{X}$ can be carried out as shown in (11). Since the fidelity term is quadratic, the solution can be easily obtained in least square estimation as follows:

$$\begin{aligned}\mathbf{X}^{k+1} = \left(\mathbf{A}^{\mathrm{H}}\mathbf{A} + \rho\mathbf{I}\right)^{-1}\left[\mathbf{A}^{\mathrm{H}}\mathbf{Y}_{\mathrm{L}} + \rho_1\left(\mathbf{Z}_1^{k+1} - \mathbf{D}_1^{k+1}\right)\right.\\ \left.+\rho_2\left(\mathbf{Z}_2^{k+1} - \mathbf{D}_2^{k+1}\right)\right].\end{aligned} \tag{26}$$

where $\mathbf{X}^{k+1}$ represents the intended result. In (26), to recover SAR image $\mathbf{X}$, the weights ρ_1 and ρ_2 are also need to be fixed, which are equivalent to the regularization parameters λ_1, λ_2 and λ_2/ρ as in the original problem in (8). Without accurate weight estimations, the solution can not satisfy the final intention of target recognition.

In this article, we consider all the weights or parameters in the above-mentioned feature quantization to be the hyperparameters, and simultaneously, the feature quantization is developed within a neural network, so that both the features and hyperparameters are overall learnable. The unfolding method [54], [55], [56] is embedded in the proposed MAFNet to nonlinearly fit the hyperparameters of the algorithmic models through the training process, and the processing flowchart of deep unfolding feature quantization is depicted in Fig. 6. Since the unfolding method is capable of replacing the artificial parameter tuning with a hyperparametric subnetwork,

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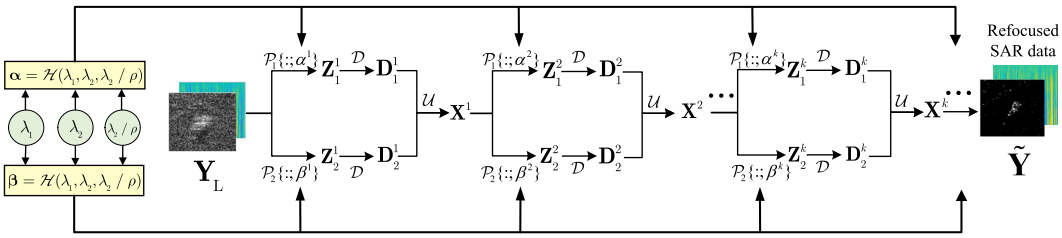


Fig. 6. Diagram of deep unfolding feature quantization.

the flexibility of the intended algorithm can be improved. In such cases, a hyperparametric subnetwork $\mathcal{H}$ will be designed to learn the hyperparameters of the regularization parameter or the weights.

According to the computations of the intended variables in Section IV-A1, the solutions of $\mathbf{Z}_1^{k+1}$, $\mathbf{D}_1^{k+1}$, $\mathbf{Z}_2^{k+1}$, $\mathbf{D}_2^{k+1}$, and $\mathbf{X}^{k+1}$ in (17), (18), (20), (22), and (26) can be updated in the unfolding manner as follows:

$$\begin{aligned}
\mathbf{Z}_1^{k+1} &= \mathcal{P}_1\left\{\mathbf{Y}_{\mathrm{L}}; \alpha^{k+1}\right\} \\
\mathbf{Z}_2^{k+1} &= \mathcal{P}_2\left\{\mathbf{X}^k, \mathbf{D}_2^k; \beta^{k+1}\right\} \\
\mathbf{D}^{k+1} &= \mathcal{D}\left\{\mathbf{D}_1^k, \mathbf{D}_2^k, \mathbf{X}^k, \mathbf{Z}_1^{k+1}, \mathbf{Z}_2^{k+1}\right\} \\
\mathbf{X}^{k+1} &= \mathcal{S}\left\{\mathbf{Y}_{\mathrm{L}}, \mathbf{Z}_1^{k+1}, \mathbf{Z}_2^{k+1}, \mathbf{D}_1^{k+1}, \mathbf{D}_2^{k+1}; \rho_1, \rho_2\right\}
\end{aligned} \tag{27}$$

where $\mathcal{P}_1\{\cdot; \alpha^{k+1}\}$ denotes the proximal mapping for IE norm and $\mathcal{P}_2\{\cdot; \beta^{k+1}\}$ denotes the proximal mapping for the ℓ_1 norm. In (27), α^{k+1} and β^{k+1} are learnable hyperparameters formulated by the hyperparametric subnetwork $\mathcal{H}$, where α^{k+1} is the output of λ_1 via $\mathcal{H}$, and β^{k+1} is the output of the combination of λ_2 and λ_2/ρ via $\mathcal{H}$. Moreover, $\mathcal{S}\{\cdot; \rho_1, \rho_2\}$ in (27) is a process of solving closed-form solution of the high resolution SAR imagery as in (26), and $\mathcal{D}\{\cdot\}$ is an analytical procedure for updating dual variables.

Conventionally, the regularization parameter or the weights are manually set according to personal experiences to balance multiple features to be enhanced [57], [58]. In this article, λ_1 and λ_2 are the parameters of the IE norm and ℓ_1 norm regularization, respectively. To avoid manual tuning of the parameters, a novel deep unfolding neural network is developed, where the parameters are learned and adjusted via a hyperparametric subnetwork $\mathcal{H}$. It is designed in a feed-forward subnetwork with three 1-D convolutional layers [54], where the first two activation functions adopt the ReLU function and the third layer employs Softplus function [59]. The kernel size of each convolutional layer is 1×1. It can realize the information exchange between channels and the addition of nonlinear features while keeping the scale of the feature map unchanged. The subnetwork $\mathcal{H}$ balances two regularization by learning the factors $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ as follows:

$$[\boldsymbol{\alpha}, \boldsymbol{\beta}] = \mathcal{H}(\lambda_1, \lambda_2, \lambda_2/\rho) \tag{28}$$

where $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ stand for the parametric matrices of the entropy norm and ℓ_1 norm regularization, respectively.

Under the process of deep unfolding feature quantization, the regularization parameters of the focusing and sparse features can be adjusted automatically to balance the multifeature quantization. As a result, the intended network is capable of accommodating the dataset contaminated by defocusing and interfered by clutter and noise with a satisfied accuracy.

B. Recognition Module

In the focusing module, both the focusing and sparse features are quantized. Also, the regularization parameters are learned automatically within a hyperparametric subnetwork. Then, a recognition module is designed, in which the quantized features are utilized to classify the targets for an accurate recognition. The architecture of the recognition module is a subnetwork $\mathcal{R}$, which is a classifier that consists of three fully connected layers. Each of the layer is activated by using LeakyReLU to produce multiple categories of the recognition result. The loss function utilized for the recognition is commonly the KL divergence. The KL divergence between two different categories is composed of an information entropy and a cross entropy. Since the information entropy of the same dataset is constant, the cross entropy is utilized to measure the difference between the multiple categories.

The smaller the cross-entropy value, the closer the probability distribution. Therefore, the prediction category corresponding to the minimum cross entropy is the result of recognition. The recognition loss function in the subnetwork $\mathcal{R}$ can be written as follows:

$$\mathcal{L}_C = -\sum_{j=1}^{10} p(x_j)\log\left(q(x_j)\right) = H\left(\mathcal{R}\{\hat{\mathbf{Y}}\}, \text{label}\right) \tag{29}$$

where $H(p(x))$ denotes the information entropy of the probability distribution p and $H(p, q)$ denotes the cross entropy between p and q. In (29), $\mathcal{R}\{\hat{\mathbf{Y}}\}$ is the predicted label, and the term "label" is the true label. With the weighing coefficients γ_1 and γ_2 derived by random sampling from a Gaussian distribution with a normalized interval of 0 to 1, the global loss function of the MAFNet can be given as follows:

$$\mathcal{L}_{\text{global}} = \gamma_1 \left\|\hat{\mathbf{Y}} - \mathbf{Y}_{\mathrm{H}}\right\|_{\mathrm{F}}^2 - \gamma_2\, H\left(\mathcal{R}\{\hat{\mathbf{Y}}\}, \text{label}\right). \tag{30}$$

By minimizing the global loss in (30), the network can simultaneously refocus the SAR imagery and identify the target accurately.

V. Processing Flowchart

The processing flowchart of the proposed MAFNet is demonstrated in Fig. 7. It can be divided into a focusing module and a recognition module. To effectively distinguish the focusing module and recognition module, the focusing module

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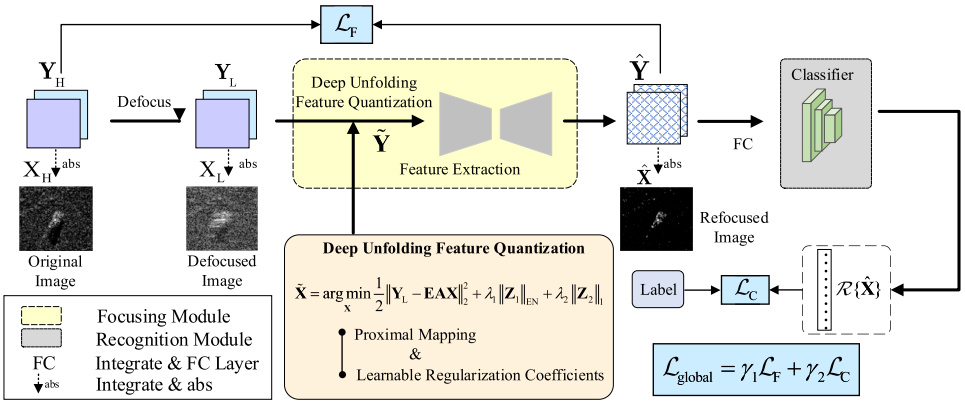


Fig. 7. Processing flowchart of the proposed MAFNet.

is represented by a yellow dashed box, and the recognition module is represented by a gray dashed box. The focusing module consists of an hyperparametric subnetwork and a U-shaped subnetwork. The architecture of the recognition module is a subnetwork $\mathcal{R}$, which is a classifier that consists of three fully connected layers.

The reason why the defocusing process is embedded in MAFNet is that the data-driven network requires learning the deep mechanism of the data from a dataset with a sufficient amount of data and diversity of samples to nonlinearly fit the network for an error-resistant SAR recognition. To this end, the defocusing process is introduced to simulate SAR data with different degrees of phase error, and a supplementary dataset is constructed. The specific defocused process is to add a quadratic distribution multiplicative phase errors of different radians in the measured data. In such a case, the phase errors present in the SAR data can be removed through network training. In order to achieve focusing and sparse feature enhancement in a "decomposition–coordination" manner, the defocused SAR data $\mathbf{Y}_{\mathrm{L}}$ are input into the deep multifeature quantization, and the output data are $\tilde{\mathbf{Y}}$, where $\tilde{\mathbf{Y}} = h(\mathbf{Y}_{\mathrm{L}})$, and $h(\cdot)$ denotes the function of deep multifeature quantization. $\tilde{\mathbf{Y}}$ is subsequently the input of the U-shaped subnetwork.

The focusing module is mainly composed of the deep unfolding feature quantization and the U-shaped subnetwork. In order to achieve feature extraction, $\tilde{\mathbf{Y}}$ gives as the input to the U-shaped subnetwork. In such a case, the sparse and refocused data $\hat{\mathbf{Y}}$ can be obtained by back-propagating the focusing loss function $\mathcal{L}_{\mathrm{F}}$ formed by $\hat{\mathbf{Y}}$ and $\mathbf{Y}_{\mathrm{H}}$, where $\hat{\mathbf{Y}} = s(\tilde{\mathbf{Y}})$, and $s(\cdot)$ is the response function of the U-shaped subnetwork that denotes the prediction mapping from the recovered features to the input data.

The recognition module is a classifier connected after the U-shaped subnetwork. The input of the classifier is the output data $\hat{\mathbf{Y}}$, and the output of the classifier is the recognition result $\mathcal{R}\{\hat{\mathbf{Y}}\}$. In such a case, the error-resistant recognition result can be obtained by back-propagating the multivariate loss function formulated in (30), which consists of the focusing loss function $\mathcal{L}_{\mathrm{F}}$ in (25) and the cross-entropy loss function $\mathcal{L}_{\mathrm{C}}$ in (29).

Particularly, the parameters of two regularizations in the deep unfolding feature quantization are learned automatically via the hyperparametric subnetwork. The advantages are twofold. On the one hand, the hyperparameters of the algorithmic model are nonlinearly fit through the training process, which improves the robustness of the recognition even when the echoed data are defocused and noised. On the other hand, the learnable parameters unfold the conventional iteration of the feature quantization into a more flexible form, which improves the interpretability of the network.

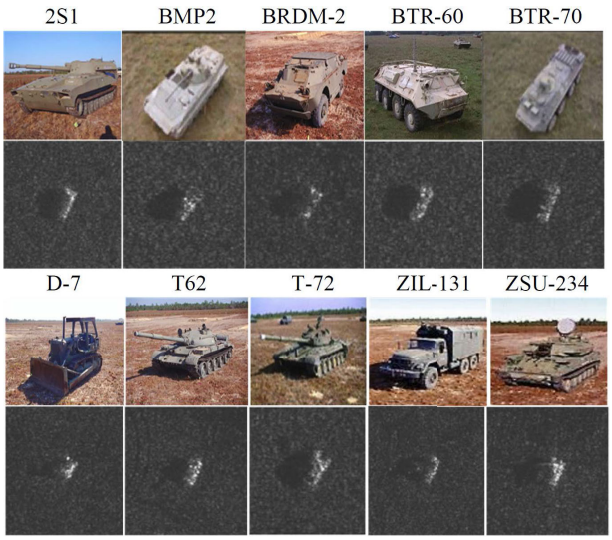


Fig. 8. Ten categories of ground stationary targets in the MSTAR dataset.

VI. Experiments

A. Experimental Setup

1) Dataset: To evaluate the performance of the proposed MAFNet, the canonical single-polarization MSTAR dataset is employed for the SAR target recognition. There are ten targets in the dataset, which are depicted in Fig. 8. In the second and fourth rows in Fig. 8, the SAR images are the truly measured data without a simulated defocusing process.

The experimental data are obtained from the measured SAR data of stationary targets published by the Moving and Stationary Target Acquisition and Recognition (MSTAR) Program supported by the Air Force Research Laboratory (AFRL) and Defense Advanced Research Projects Agency (DARPA) [60].

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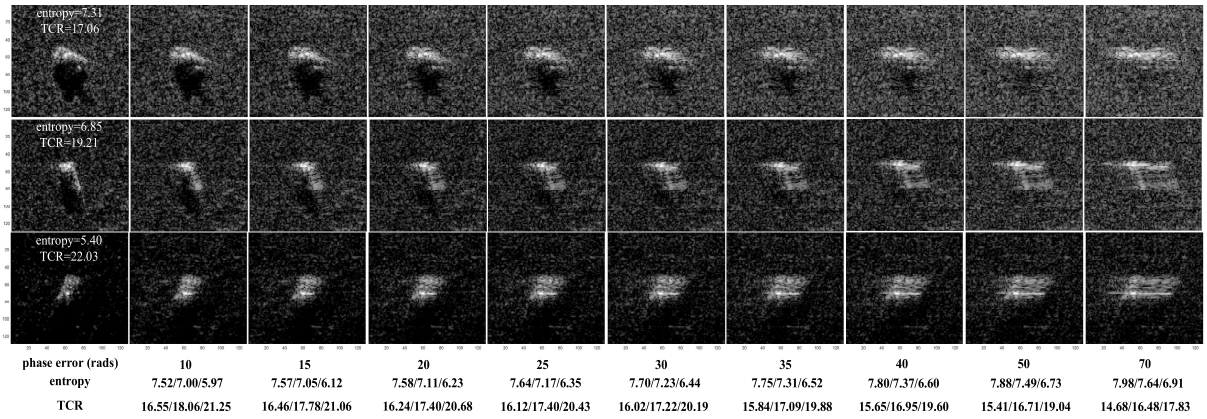


Fig. 9. SAR targets with different phase errors. Three rows of SAR images show the visualization results of three kinds of stationary ground targets with different radii of phase error.

TABLE I
SETUP OF THE MSTAR DATASET

class	Training Sets		Testing Sets	
	Size(height*width)	Pitch Angle	Size(height*width)	Pitch Angle
2S1	158 × 158	17°	158 × 158	15°
BMP-2	128 × 128	17°	128 × 128	15°
BDRM-2	129 × 128	17°	129 × 128	15°
BTR-60	128 × 128	17°	128 × 128	15°
BTR-70	128 × 128	17°	128 × 128	15°
D-7	178 × 177	17°	178 × 177	15°
T-62	173 × 172	17°	128 × 128	15°
T-72	128 × 128	17°	128 × 128	15°
ZIL-131	193 × 192	17°	193 × 192	15°
ZSU-234	158 × 158	17°	158 × 158	15°

The data are collected by a wideband synthetic aperture radar (SAR) with a resolution up to 0.3 × 0.3 m, where the radar works on X band and single polarization.

2) Experimental Preparations: Details about the single-polarization MSTAR dataset are illustrated in Table I, where the size of the SAR images is not uniform. Therefore, all the SAR images are cropped to 128 * 128 resolution cells in the preparation stage. The use of 15° and 17° pitch angle has little effect on the imaging results. The imaging results with different pitch angles were used as testing data and training data, respectively, just to separate the testing data and training data, thoroughly. Then, the data augmentation of the training set is performed by simulating SAR images with quadratic distribution multiplicative phase errors of 10, 15, and 20 rad, respectively, and the data is randomly arranged and combined to obtain a training dataset containing both defocused and focused data. By combining the original training set with the augmented data, a larger training set are obtained. The resulting training set has 6772 samples in total. The testing data utilize the data with no phase error and the defocused data with 0, 10, 15, 20, 25-, 30-, 35-, 40-, 50-, and 70-rad phase error. The defocused process is to add a quadratic distribution of multiplicative phase errors of different radians in the measured data. Specifically, the number of samples for the training set and testing set is allocated in a ratio of 70%–30%. SAR images with different phase errors are shown in Fig. 9. It is noteworthy that it is the motion error in the slant range direction that is directly related to the defocusing problem, and we only consider the recovery of the projection of the motion error in the slant range direction in the experiments [61]. The purpose of assigning these defocused samples to the training set is to familiarize the network with the motion error and enable the network to learn the potential features due to the motion errors. Since there is an implicit dimensional transformation between the echoed data and the SAR image, the Fourier transformation block is applied to transform complex-valued echoed data into the SAR image that can be used for autofocusing processing.

B. Network Ablation Study

In this section, the ablation is to study the performance of the proposed MAFNet under various conditions. In the ablation experiments, the improvement of the accuracy of recognition may be small. Even the same server is utilized for training, the experimental results may still fluctuate due to the generation of random numbers. Therefore, to make the results of comparative experiments accurately, the experimental platform is a GeForce RTX 2080 server, and the random seed is fixed in the experiments.

1) Result of Target Autofocusing: The result for SAR data with different phase errors is shown in Fig. 9. The specific defocused process is to add a quadratic distribution multiplicative phase errors of different radians in the measured data. When the phase error reaches 70 rad, the features of the target cannot be distinguished. Therefore, the SAR image with 70-rad phase error is utilized for target autofocusing, and the result is shown in Fig. 10. Specifically, Fig. 10(a) shows a truly measured data acquired in a single polarization MSTAR dataset, with inevitable additive noise due to the ground clutter and other complicated electromagnetic environment interferences. Fig. 10(b) shows the SAR image with simulated quadratic phase error of 70 radians. Visually, the target in Fig. 10(b) is smeared and its outline is indistinguishable. The result of the CCNN is provided in Fig. 10(c), where it is refocused compared with Fig. 10(b). However, in Fig. 10(c), a horizontal line due to defocus can still be faintly seen in the middle of the vehicle target, and the background noise is not effectively suppressed. This is mainly due to CCNN is only a data-driven network, which relies on the similarity between the predicted SAR image and the focused SAR image

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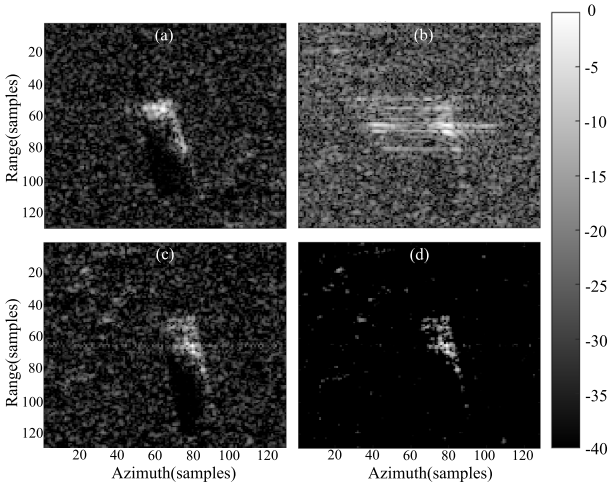


Fig. 10. Result of target refocusing. (a) Ground truth. (b) Defocused target with 70-rad phase error. (c) Conventional CCNN. (d) Proposed MAFNet.

to guide the network training. In other words, the mathematical and physical interpretability of CCNN is weak. Then, the generalization ability of the CCNN is also limited, and it would perform poorly under large phase errors.

The result of the proposed MAFNet is provided in Fig. 10(d), where it is refocused compared with Fig. 10(b) and (c). In Fig. 10(d), the defocus effect is almost eliminated. It is obvious that the background noise is highly suppressed and the target is almost fully refocused, thus, the target can be easily distinguished. This is mainly due to the MAFNet is a model-driven neural network, where the deep unfolding is used to achieve a hybrid of the model-driven and data-driven framework. Specifically, the IE is utilized to quantify the focusing feature to guide the MAFNet framework to autofocus the SAR images, and the ℓ_1 norm regularization optimization is incorporated to extract the sparse feature of the SAR imagery to suppress background noise cooperatively in the MAFNet. As a result, in Fig. 10(d), the proposed MAFNet significantly improves the SAR image refocusing and noise suppression compared with the conventional CCNN result shown in Fig. 10(c).

In terms of the quantitative aspect, the IE and target-to-clutter ratio (TCR) are provided. IE is used to evaluate the focusing quality of SAR images. The smaller the IE, the higher the focusing quality. TCR is used to evaluate the signal-to-noise ratio of the image. The higher the TCR value, the higher the contrast of the image. In Table II, IE of Fig. 10(c) is smaller than that of Fig. 10(b), and the TCR is larger than that of Fig. 10(b). Compared with those in Fig. 10(b) and the target in Fig. 10(c) recovered by conventional CCNN, the result of the model-driven and data-driven MAFNet in Fig. 10(d) performs with the IE of 4.1639 and the TCR of 31.2777. Therfore, in Fig. 10, the results produced by the CCNN and proposed MAFNet are both reasonable. In addition, the IE of the proposed MAFNet in Fig. 10(d) is the smallest, and the TCR is the largest. It also means that the refocused and denoised effect of the MAFNet is better than the CCNN. Because of the model-driven and data-driven are hybridized in the MAFNet, the focusing feature and sparse feature are

TABLE II
QUALITY METRICS FOR TARGET REFOCUSING IN FIG. 10

Data	IE	TCR(dB)
(a) Ground Truth	6.8532	19.2092
(b) Defocused data	8.1592	13.5549
(c) CCNN	7.1177	17.8580
(d) MAFNet	**4.1639**	**31.2777**

enhanced to simultaneously obtain the refocused and denoised SAR image and the high-target recognition accuracy.

Given multiple MSTAR targets, the output feature maps of MAFNet are illustrated in Fig. 11. In Fig. 11(a) and (b), the contour features in the low level are extracted via the first cross-convolutional layer. Since the defocusing changes the contour features of the target, the contour features in Fig. 11(b) are stretched horizontally. In Fig. 11(c) and (d), the features based on key points are extracted via the second cross-convolutional layer. In addition, the defocused features based on key points in Fig. 11(d) have lost their spot-like features visually. In Fig. 11(e) and (f), the deep refinement features are extracted via the third cross-convolutional layer. Similarly, the defocused deep refinement features also appear to be stretched to enlarge the feature region. To sum up, the feature maps of the targets are blurred due to defocusing, which proves the necessity of SAR target autofocusing.

Apart from the results of visualization, we also provide some quantitative comparisons of different methods. Fig. 12 is presented to compare the IE value, i.e., IE of defocused data, focused only data, sparse only data, and refocused (focused and sparse) data, respectively. In Fig. 12, the phase error is simulated to 30 rad. The focusing quality of the proposed algorithm is better than that of the other algorithms, since the IE is the smallest. In addition, the quality of the SAR imagery after refocusing is better than other algorithms, which proves that the improvement of recognition accuracy is related to the improvement of focusing quality in the focusing module.

2) Result of Target Recognition: When contaminated by large phase errors, the average recognition results of different methods are recorded in Table III. The best results are highlighted in bold. CNN denotes the preprocessing model ResNet18 [19], and the focusing module of FC-CCNN is based on a fully convolutional architecture.

The focusing module of U-CCNN is a U-shaped subnetwork depicted in Fig. 5. It should be noted that the loss functions in CNN, FC-CCNN, and U-CCNN [62] are only cross entropy. Based on FC-CCNN and U-CCNN, the IE is introduced in the focusing loss function of FCE-CCNN and UE-CCNN. Based on FCE-CCNN and UE-CCNN, the ADMM algorithm is utilized in FCA-CCNN and MAFNet. In order to evaluate the multivariate loss function, two networks with a separate focusing network and recognition network are designed in the experiment. UA1-CCNN uses a U-shaped network for focusing, and a conventional CCNN for recognition. UA2-CCNN adopts U-shaped networks for both focusing and recognition.

In Table III, MAFNet outperforms other methods when the phase error of SAR data is 25, 30, 35, 40, and 50 rad. In the most extreme case of 70-rad phase error, the accuracy

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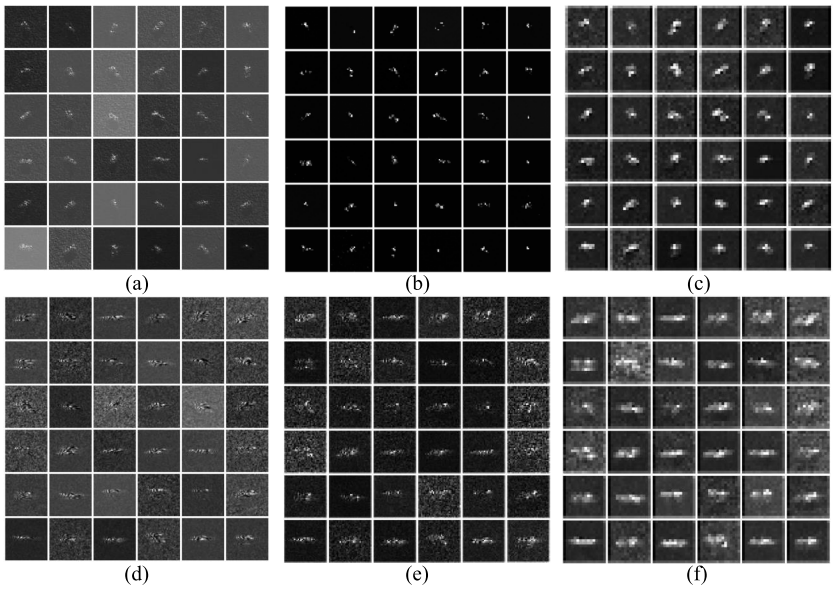


Fig. 11. Output feature maps of different layers of the MAFNet when given an MSTAR target. (a) Focused feature maps of cross convolution #1. (b) Defocused feature maps of cross convolution #1. (c) Focused feature maps of cross convolution #2. (d) Defocused feature maps of cross convolution #2. (e) Focused feature maps of cross convolution #3. (f) Defocused feature maps of cross convolution #3.

TABLE III
RESULTS (ACCURACY) FOR NETWORKS TESTED ON DATA WITH DIFFERENT PHASE ERRORS USING RANDOM SEEDS

Network	(phase error)	CNN	FC-CCNN	FCE-CCNN	FCA-CCNN	U-CCNN	UE-CCNN	UA1-CCNN	UA2-CCNN	**MAFNet**
accuracy	25 rad	0.8252	0.9328	0.9563	0.9769	0.9852	0.9872	0.9847	0.9843	**0.9889**
	30 rad	0.7677	0.8648	0.9233	0.9406	0.9563	0.9670	0.9604	0.9678	**0.9761**
	35 rad	0.6976	0.7906	0.8520	0.8681	0.8937	0.9138	0.9060	0.9122	**0.9312**
	40 rad	0.6191	0.7143	0.7646	0.7857	0.7898	0.8244	0.8026	0.8157	**0.8623**
	50 rad	0.4786	0.6080	0.6352	0.6323	0.5709	0.6121	0.5969	0.6068	**0.6814**
	70 rad	0.3323	0.4683	**0.4687**	0.4534	0.3108	0.3355	0.2840	0.3215	0.4126

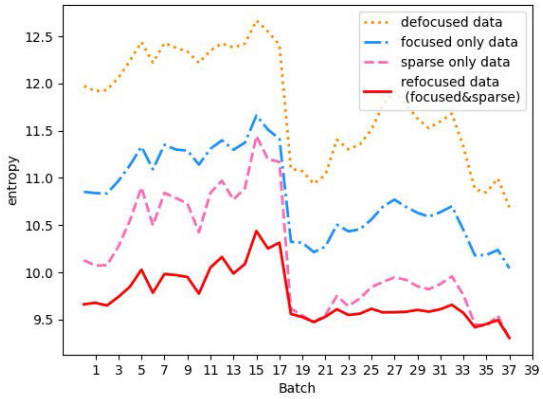


Fig. 12. IE of different data.

of the networks using the U-shaped architecture is lower than that using the fully convolutional architecture. Moreover, the accuracy of MAFNet is higher than other methods using the U-shaped architecture. Furthermore, the accuracy of MAFNet is worse than that of using the fully convolutional architecture. The reason is that the existence of the max-pooling layer causes the loss of some spatial features. However, it is still better than other networks using the U-shaped architecture, which confirms the robustness and the superiority of MAFNet.

TABLE IV
CONFUSION MATRIX OF MAFNet WITH NONE PHASE ERROR

Class	2S1	BMP-2	BDRM-2	BTR-60	BTR-70	D-7	T-62	T-72	ZIL-131	ZSU-234	PCC(%)
2S1	**269**	0	2	1	1	0	1	0	0	0	98.18
BMP-2	0	**196**	0	0	0	0	0	0	0	0	100
BDRM-2	0	0	**274**	0	0	0	0	0	0	0	100
BTR-60	0	0	0	**193**	2	0	0	0	0	0	98.97
BTR-70	0	0	0	0	**196**	0	0	0	0	0	100
D-7	0	0	0	0	0	**294**	0	0	0	0	100
T-62	0	0	0	0	0	0	**273**	0	0	0	100
T-72	0	0	0	0	0	0	0	**196**	0	0	100
ZIL-131	0	1	0	0	0	0	0	0	**273**	0	99.64
ZSU-234	0	0	0	0	0	3	0	1	0	**270**	98.54

TABLE V
CONFUSION MATRIX OF MAFNet WITH PHASE ERROR OF 10 rad

Class	2S1	BMP-2	BDRM-2	BTR-60	BTR-70	D-7	T-62	T-72	ZIL-131	ZSU-234	PCC(%)
2S1	**268**	1	2	1	1	0	1	0	0	0	97.81
BMP-2	0	**196**	0	0	0	0	0	8	0	0	100
BDRM-2	0	0	**274**	0	0	0	0	0	0	0	100
BTR-60	0	0	0	**193**	2	0	0	0	0	0	98.97
BTR-70	0	0	0	0	**196**	0	0	0	0	0	100
D-7	0	0	0	0	0	**274**	0	0	0	0	100
T-62	0	0	0	0	0	1	**272**	0	0	0	99.63
T-72	0	0	0	1	0	0	0	**195**	0	0	99.49
ZIL-131	0	1	0	0	0	0	0	0	**273**	0	99.64
ZSU-234	0	0	0	0	0	3	0	1	0	**270**	98.54

The confusion matrix of the MAFNet without phase error is shown in Table IV, while the confusion matrix with phase errors of 10, 15, and 20 are shown in Table V–VII, respectively. Confusion matrix [63] is a quantitative index

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TABLE VI
CONFUSION MATRIX OF MAFNet WITH PHASE ERROR OF 15 rad

Class	2S1	BMP-2	BDRM-2	BTR-60	BTR-70	D-7	T-62	T-72	ZIL-131	ZSU-234	PCC(%)
2S1	**269**	0	2	1	1	0	1	0	0	0	98.18
BMP-2	0	**195**	1	0	0	0	0	0	0	0	99.49
BDRM-2	0	0	**274**	0	0	0	0	0	0	0	100
BTR-60	0	1	0	**193**	1	0	0	0	0	0	98.97
BTR-70	0	0	0	0	**196**	0	0	0	0	0	100
D-7	0	0	0	0	0	**274**	0	0	0	0	100
T-62	1	0	0	0	0	1	**271**	1	0	0	99.27
T-72	0	1	0	0	0	0	0	**195**	0	0	99.49
ZIL-131	0	0	0	0	0	0	0	0	**273**	1	99.64
ZSU-234	0	0	0	0	0	5	0	1	0	**268**	97.81

TABLE VII
CONFUSION MATRIX OF MAFNet WITH PHASE ERROR OF 20 rad

Class	2S1	BMP-2	BDRM-2	BTR-60	BTR-70	D-7	T-62	T-72	ZIL-131	ZSU-234	PCC(%)
2S1	**269**	0	2	2	0	0	1	0	0	0	98.18
BMP-2	0	**195**	0	0	1	0	0	0	0	0	99.49
BDRM-2	0	1	**273**	0	0	0	0	0	0	0	99.64
BTR-60	0	2	0	**191**	2	0	0	0	0	0	97.95
BTR-70	1	1	0	0	**194**	0	0	0	0	0	98.98
D-7	0	0	0	0	0	**274**	0	0	0	0	100
T-62	0	0	0	0	0	2	**270**	1	0	0	98.90
T-72	0	1	0	0	0	0	0	**195**	0	0	99.49
ZIL-131	0	0	0	0	0	0	0	0	**273**	1	99.64
ZSU-234	0	0	0	0	0	5	0	1	0	**268**	97.81

TABLE VIII
RESULTS FOR MAFNet TESTED ON DATA WITH DIFFERENT PHASE ERRORS USING RANDOM SEEDS

Phase Error	Accuracy	F_1 Score	Precision	Recall
No Error	0.9951	0.9950	0.9953	0.9948
10 rad	0.9938	0.9937	0.9941	0.9934
15 rad	0.9926	0.9926	0.9929	0.9925
20 rad	0.9901	0.9898	0.9901	0.9896

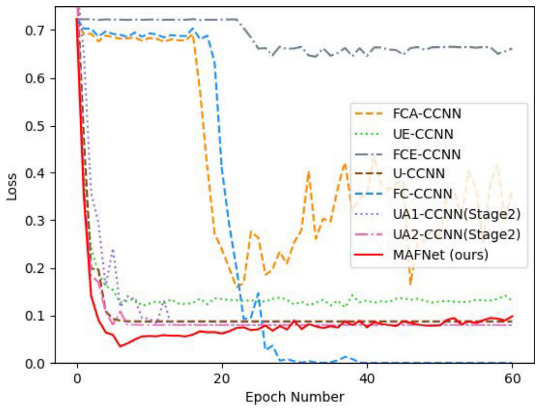


Fig. 13. Normalized training loss of different methods.

to evaluate the results of recognition, which records the predicted results in the testing process of each category. Based on the amount of real and predicted samples in the confusion matrix, recognition indexes, such as accuracy, F_1 score, precision, and recall rate are sorted out in Table VIII. It can be noted from the confusion matrix that the probability of correct classification (PCC) of the MAFNet with phase errors of 0, 10, 15, and 20 rad is all more than 99%. Particularly, by averaging the result, the MAFNet with no phase error achieves the PCC of 99.497%. Generally, the proposed MAFNet can achieve high accuracy in relatively small phase errors. In Table VIII, the high recognition index values confirm the accuracy and robustness of the MAFNet.

C. Convergence of the Networks

Fig. 13 records the normalized focusing loss distribution of different networks within 60 epochs, and utilizes lines in different styles to represent the convergence characteristics of different networks. UA1-CCNN (Stage2) and UA2-CCNN (Stage2) respectively represent the convergence curves of the recognition stage in the two-stage networks UA1-CCNN and UA2-CCNN. The following characteristics can be found through visual perception. It can be noted that in the convergence curves, the convergence efficiency of networks using the U-shaped architecture is superior than that using the fully convolutional architecture. FCA-CCNN does not converge within 60 epochs. In addition, the average loss value of MAFNet is smaller than that of other networks using the U-shaped architecture. It proves the efficient convergence of MAFNet.

VII. CONCLUSION

In this article, the MAFNet is proposed to maintain a robust recognition accuracy even in the presence of nonsystematic phase error contamination. The minimum IE is used to evaluate and compensate the azimuthal phase error to achieve image refocusing. The entropy is statistically based, and the minimum entropy cannot be simply concluded as representing the contrast of an image. Two key features of focusing and sparse features are quantized and solved in closed-form solutions, so that the SAR image can be interpreted accurately, and the accuracy of target recognition can be improved even the dataset is defocused and noised. Specifically, the deep unfolding technique is embedded to nonlinearly fit the hyperparameters of the algorithmic models in the training process. It improves the quality of refocused data, the accuracy of SAR recognition, and the interpretability of the network, accordingly and simultaneously. The canonical single polarization MASTAR dataset is employed for the verification. Under the single polarization, various comparisons with conventions are performed to reveal the superiority of the MAFNet algorithm in both qualitative and quantitative aspects. The complex-valued image and continuous phase history are considered, which improves the applicability of the proposed algorithm. The feasibility of the proposed algorithm for the multipolarizations is under investigation. For the study of generalization of polarization and complicated backgrounds, a dataset of different polarization and complicated backgrounds will be needed. The useful data will be collected, and further experiments will be investigated in our future work.

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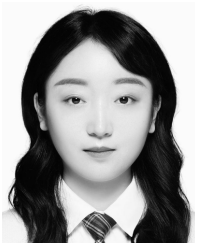
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